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Early Percutaneous Cryoneurolysis for Pain Control After Rib Fracture for Older Patients: A Prospective Randomized Clinical Trial

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Short Title: Cryoneurolysis for Rib Fracture Pain

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Background: Rib fractures are common and multimodal pain control bundles are essential to optimal care. Minimizing therapeutic side effects is important among older adults. The purpose of this study is to determine whether addition of CT-guided percutaneous cryoneurolysis (CT-PCN) to existing multimodal pain control bundles reduces pain associated with rib fractures in older adults.

Study Design: A prospective, randomized clinical trial assessed addition of CT-PCN to a multimodal rib fracture pain control bundle (Standard of Care [SoC]) for injured adults ≥65 years. Bluntly-injured patients with fractured ribs 3-9 were randomized ≤72 hours from admission to CT-PCN and SoC, or SoC alone. Primary outcome was highest pain score within 24 hours of discharge with deep inspiration. Secondary outcomes included hospital length of stay, rib-specific readmissions, intervention-associated adverse events, morphine milligram equivalent (MME) use, short-form (SF)-12, and McGill pain scores, assessed at discharge and 1-, 3-, and 12-month intervals post-discharge.

Results: 110 patients were included (56 intervention, 54 SoC). No difference in highest pain score with deep inspiration (intervention mean 5.6(SD 2.4) vs SoC mean 5.9(SD 2.4); P=0.49) was within 24 hours of discharge between groups and there was no difference in MME use on day-of-discharge (intervention median 24(IQR 6-45) vs SoC median 15(IQR 4-34); P=0.29). Among 77 participants completing 1-month follow-up (39 intervention, 38 SoC), mean numeric pain score was 1 point worse among intervention patients (intervention mean 2.7(SD 2.4) vs SoC mean 1.7(SD 1.8); P=0.04) and mean sensory pain score was 3.1 points higher (95% CI1.4- 4.9; P<0.001).

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Conclusions: CT-PCN did not decrease pain associated with rib fractures and may be associated with worse physical quality and emotional response to pain within one month after hospital discharge.

Keywords: pain, rib fracture, disability, nerve

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Background:

Rib fractures are common injuries sustained after trauma. Adverse outcomes associated with rib fractures rise with age starting as early as the fifth decade of life and increases significantly in patients over ≥65 years (1-3). Just as rib fracture morphology exists along a spectrum, so do therapies, ranging from oral and intravenous (IV) pain medications, locoregional analgesia, and surgical stabilization of rib fractures (SSRF) (4-7). An optimized, individualized, multimodal care bundle takes advantage of each pain modality while minimizing side effects. Side effect mitigation is especially important for older adults, where medical co-morbidities, lower volumes of distribution, and impaired clearance can quickly compound, causing any therapy to exacerbate the initial traumatic injury. A number of institutions, including our own, have implemented evidence-based, multidisciplinary care pathways to improve clinical outcomes in patients with rib fractures (8-11). These care pathways may be particularly beneficial to older patients (11, 12).

Intercostal nerve cryoneurolysis may offer a minimally invasive method to deliver long term pain control. Direct application of cold to nerves destroys nerve axons, resulting in Wallerian degeneration of the distal nerve without distorting epineural or perineurial tissue (13, 14). Preservation of peri- and epineurium provides a scaffold for peripheral nerve regeneration enabling reversible anesthesia usually lasting 3-6 months (13-15). Computed tomography (CT)-guided cryoablation of intercostal nerves has been shown to decrease chronic pain following thoracotomy (16) and case reports suggested potential benefit from cryoneurolysis in patients with rib fractures (14). These findings suggest a potential role for percutaneous, CT-guided intercostal nerve cryoablation in patients with rib fractures. Our hypothesis was addition of CT-

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guided percutaneous cryoneurolysis to a multimodal pain control bundle would reduce pain associated with rib fractures in older adults.

Methods:

We performed a prospective, randomized clinical trial at our American College of Surgeons Level 1 trauma center to assess the addition of CT-guided percutaneous cryoneurolysis to our existing multimodal rib fracture pain control bundle (Standard of Care [SoC]) for injured adults 65 years of age or older. This clinical trial was prospectively registered with clinicaltrials.gov (NCT04482582) on July 20, 2020. CONSORT reporting guidelines were adhered to and the CONSORT checklist is provided (Supplemental Digital Content 1, http://links.lww.com/JACS/A630). Bluntly injured patients with fractured ribs 3-9 were randomized within 72 hours of admission to receive either CT-guided, percutaneous cryoneurolysis and SoC, or SoC alone.

Patients undergoing SSRF were excluded (Supplemental Digital Content 2, http://links.lww.com/JACS/A631). Patients < 65 years were excluded due to a competing, in-parallel clinical trial evaluating younger patients undergoing ultrasound-guided percutaneous cryoneurolysis (NCT05330611). CT-guided cryoneurolysis was chosen for this initial study as this was the only cryoneurolysis device modality available at our institution during study conception and protocol development. Additional exclusion criteria included pregnancy, incarceration, inability to communicate or participate with investigators, inability to consent for procedure, pain score <5 with deep inspiration, radiographic evidence of metastasis to ribs, Glasgow Coma Scale [GCS] <13, rib fractures located <3cm from the transverse process, coagulopathy (international normalized ratio [INR] >1.5, platelet count <100K), or inability to tolerate procedure positioning. Patients with any contraindication specific to percutaneous

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cryoneurolysis, including cold urticaria, cryofibrinogenemia, cryoglobulinemia, paroxysmal cold hemoglobinuria, and Raynaud's disease were excluded. Rib fracture severity was allocated according to existing nomenclature using the radiology report; if a given rib was fractured in more than one location, the more severe fracture was recorded (4).

The primary outcome was highest pain score with deep inspiration (at least 500cc on incentive spirometry) within 24 hours of discharge. Pain scores were obtained by nursing staff in a trained acute care for elders (ACES) trauma unit in accordance with unit protocol. Standardization of timing of physical therapy (PT) and occupational therapy (OT) participation prior to acquisition of pain score was not performed given resource constraints. Secondary outcomes included hospital length of stay, intervention-associated adverse events, morphine milligram equivalent (MME) use, short form 12 (SF-12), and McGill pain scores (PROs) which were assessed at discharge and 1-, 3-, and 12-month intervals after discharge. PRO results were obtained by phone. A rib-specific readmission was defined as a readmission associated with delayed hemothorax or pneumothorax, need for chest wall-specific pain control, pneumonia, or empyema. Patients were randomly allocated by the research coordinators to one of two treatment groups within 72 hours of admission: i) SoC alone (Supplemental Digital Content 3, http://links.lww.com/JACS/A632) or ii) SoC with CT-guided percutaneous cryoneurolysis. Patients randomized to intervention were required to have intervention within 72 hours. Computer generated randomization was used by the clinical trials team who were uninvolved with patient care. Randomization occurred in a 1:1 ratio after study enrollment. No blinding was performed.

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Patients randomized to intervention were consented to the study procedure and evaluated for safe positioning in prone or decubitus position. The procedure was performed in an interventional radiology procedure suite with a Siemens Somatom Definition Edge scanner (Siemens Healthineers, Erlangen, Germany) using intermittent CT-guidance or CT-fluoroscopy. All interventional radiologists performing this procedure had ≥5 years of experience performing percutaneous cryoablation procedures. The procedure was performed under moderate sedation using fentanyl and midazolam. Non-contrast CT was used to identify subject anatomy and target ribs were identified in cross-sectional view. Local anesthesia was administered and an Endocare™ 1.7mm Round Ice PerCryo™ Cryoprobe (Varian Medical Systems, Palo Alto, CA) was advanced in the first treatment level and positioned inferior to the targeted rib approximately 3cm from the transverse process. Cryoneurolysis was performed with a 90-second freeze at 30% gas flow following study protocol. All accessible nerves of ribs 3-9 on the injured hemithorax were ablated, if possible. Post-treatment imaging was performed to evaluate for new or worsening pneumothorax or hemothorax.

Data for enrolled patients and screen failures was collected in REDCap. Reasons for screen failures were recorded. At 30% and 50% enrollment, interim analyses were performed and published in accordance with study protocol (17). No patient or public involvement occurred during trial design. An *a priori* sample size calculation estimated 110 participants would have 80% power to detect a 2-point difference in mean pain score between groups, assuming a standard deviation of 3 and significance level of 0.05. Differences in pain scores between groups were assessed using independent samples *t*-tests, with the exception of McGill Affective Pain Scores, which were compared using Wilcoxon rank-sum tests due to skew. Comparisons of numeric variables were performed using Wilcoxon rank-sum tests, and categorical variables

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were analyzed using Chi-Square or Fisher's Exact tests, as appropriate. Longitudinal changes in pain among intervention patients were evaluated using Wilcoxon signed-rank tests. A linear mixed-effects model with random intercept and random slope evaluated differential change in pain over time between groups. To preserve overall Type I error rate, the O'Brien-Fleming alpha spending function with two preplanned interim analyses was implemented, which provided adjusted significance boundaries. For the final analysis, statistical significance was defined as a two-sided P-value of 0.045 or less. The primary intention-to-treat analysis included all patients, regardless of whether they received the assigned treatment. Per-protocol analysis excluded intervention patients who did not ultimately undergo cryoneurolysis. Post-hoc, an additional subgroup analyses were performed among patients a) with at least 1 displaced rib fracture and b) with at least 1 rib fracture(s) in ribs 1, 2, 10, 11, or 12 on the affected side. Analyses were conducted using SAS, version 9.4 (SAS Institute Inc, North Carolina).

Written informed consent was obtained from all patients. No compensation was provided to patients for this study. Chest Wall Injury Society (CWIS) recommendations for reporting studies in chest wall injury were followed as applicable (18). Institutional review board approval was obtained for this clinical trial.

Results:

Three-hundred seventy injured patients were approached to participate between June 29, 2021 and April 15, 2024 (Figure 1). One-hundred twenty-seven patients consented to enroll. Among the 127 patients who consented, 110 were included in the study (56 intervention, 54 SoC). The remaining 17 were screen failures (N = 13), dropped prior to intervention (N = 2), dropped by the principal investigator (N = 1), or were off-protocol (N = 1) and thus excluded from the study. Common reasons for screen failure were an inability to tolerate prone positioning

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(4/13), unable to schedule cryoneurolysis within study timeframe (4/13), fractures too posterior to target (2/13), inability to follow procedural instructions (2/13), and traumatic brain injury with progression of intracranial hemorrhage (1/13). To address rate of screen failures, a protocol amendment implementing an intention-to-treat (ITT) strategy was initiated October 1, 2023. Among the 110 patients included in the study, 7 were included as intent-to-treat patients. These patients were included in the intervention group but did not receive intervention.

Patients were well-matched (Table 1). Median (IQR) age was 80(72-86) years and 59(54%) were female. All patients presented with a blunt injury, and most (N=82, 75%) presented after fall. Median (IQR) injury severity score (ISS) was 10(9-14). Median (IQR) number of rib fractures was 5(3-6) and 13 patients (12%) experienced bilateral fractures. Seventy-one (65%) patients had at least one displaced fracture and 3(3%) had a radiographic flail segment. Hemo- and pneumo-thoraces on admission were common, with 34(31%) and 25(23%) patients having these conditions, respectively. Prior to randomization, highest pain score in the first 24 hours was median 8(IQR 7-10). Median daily MME per patient in the first 24 hours after admission was 25(IQR 12-42).

There were no differences between hospitalization or discharge variables between the two groups (Table 2). Time from emergency department arrival to trial enrollment was equivalent between groups (intervention: median 1(IQR 1-1) days vs SoC: median 1(IQR 1-2) days; P=0.22). Among intervention patients, cryoneurolysis was performed within a median 31(IQR 22-55) hours from admission. Median number of intercostal nerves undergoing cryoneurolysis was 5(range: 3-6). Forty-seven (96%) patients had either 5(N=40) or 6(N=7) levels frozen. Twenty-five (51%) patients had left-sided cryoneurolysis, 21(43%) had right-sided cryoneurolysis, and 3(6%) had bilateral cryoneurolysis. After intervention, a subsequent nerve

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block was required in 7(13%) intervention patients compared to 19(35%) SoC patients (P=0.005). Two SoC patients required a chest tube for progressing hemopneumothoraces as did two intervention patients; none of these were identified as procedure related. No patients required intubation. Only 2(4%) SoC patients experienced pneumonia during their hospitalization.

Median hospital length of stay was 4(IQR 3-7) among intervention patients and 6(IQR 3-7) among SoC patients (P=0.77).There was no difference between intervention patients and SoC patients in hospital disposition (intervention 34(61%) vs SoC 29(54%) discharged home; P=0.46). There was no difference in highest pain score with deep inspiration (intervention mean 5.6(SD 2.4) vs SoC mean 5.9(SD 2.4); P=0.49) in the last 24 hours before discharge between groups (Figure 2). There was no difference in MME use on day of discharge between groups (intervention median 24(IQR 6-45) vs SoC median 15(IQR 4-34); P=0.29) (Figure 3). Results were consistent in the per-protocol analysis. Non-steroidal anti-inflammatory drug (NSAID) use was greater among intervention patients upon discharge (intervention 11(20%) vs SoC 3(6%); P=0.03).

Among 77 participants who completed 1-month follow-up (39 intervention, 38 SoC), mean numeric pain score at 1-month was significantly different between groups (intervention mean 2.7(SD 2.4) vs SOC mean: 1.7(SD 1.8); P=0.04) (Table 3). However, the per-protocol analysis did not find a significant difference in mean numeric pain score between groups (p = 0.07. More intervention participants were still on opioid pain medications 1-month compared to SoC (32% vs 12%; P=0.01). Intervention patients were more likely to still require gabapentin compared to SoC (45% vs 12%; P<0.001). At 1-month, SF-12 PCS and MCS scores were not significantly different between groups. However, intervention participants reported a mean

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sensory pain score that was 3.1 points higher than the SoC group (95% CI:1.4-4.9; P<0.001) and significantly higher affective scores (median(IQR) = 1(0-2) vs. 0(0-0); P = 0.01).

Among 72 participants who completed 3-month follow-up (35 intervention, 37 SOC), mean numeric pain at 3-months was not significantly different between groups (intervention mean (SD):1.4(1.8), SOC mean (SD):0.9(1.3); difference (95% CI):0.5(-0.2-1.3]), P=0.15). Lidocaine patch use remained higher among intervention patients compared to SoC patients (intervention 14(26%) vs SoC 3(6%);P=0.008). There were no differences in McGill pain scores or SF-12 at 3-month follow-up.

Sixty-two participants completed 1-year follow-up (30 intervention, 32 SOC). Only one intervention patient was still taking opioid pain medication at 1-year. There were no differences in outcomes between groups at 1-year. By 1 year after initial hospitalization, only one patient experienced a rib-specific readmission in the SoC arm. Six patients (three in each arm) died within 1 year of hospital discharge. All deaths occurred during the follow-up period and none were attributed to the intervention or to traumatic rib fractures. Length of stay was significantly higher among patients lost to follow-up compared to patients who completed all follow-ups (median () (IQR) 5(3-7) vs. 4(3-6); P=0.04) .

Sub-group analysis of the 71 patients (32 intervention, 39 SoC) with ≥1 displaced rib fracture similarly did not identify any difference between groups with respect to highest pain score in last 24 hours before discharge (intervention mean 6.1(SD 2.2) vs SoC 5.6(SD 2.5), P=0.46) or MME use on day of discharge (intervention median 29.2(IQR 8-45) vs SoC median 15.0(IQR 4.1-30.0),P=0.07). Compared to SoC, at one month mean PPI score was 1.3 points higher (intervention mean 3.0(SD 2.6) vs SoC mean 1.6(SD 1.7); P=0.03), mean sensory pain score was 4.2 points higher (intervention mean 5.7(SD 5.4) vs SoC mean 1.5(SD 2.1); P<0.001),

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and distribution of affective pain scores was higher among intervention participants (intervention median 1(IQR 0-2) vs SoC median 0(IQR 0-0); P=0.003). Daily opioid use in MME was also higher among intervention patients at one month compared to SoC patients (intervention median 0(IQR 0-6.6) vs SoC median 0(IQR 0-0); P=0.02). These differences between treatment arms in the subgroup analysis of patients with ≥1 displaced fracture resolved at follow-up at 3-months and one year. Subgroup analysis of the 51 patients (18 intervention and 33 SoC) with additional fractures of ribs 1,2,10,11, and 12 demonstrated no difference in either highest pain score with deep inspiration at discharge nor any of the McGill Pain Score components at 1-month follow-up.

Discussion:

In contrast to several recent studies demonstrating benefit of intercostal cryoneurolysis in patients with rib fractures (19-22), we found no benefit to percutaneous cryoneurolysis in reducing pain associated with rib fractures. There was no difference in highest pain scores with deep inspiration within the last 24 hours before discharge or MME use at discharge. Patients receiving intercostal cryoneurolysis were more likely to require NSAIDs, gabapentin, lidocaine patches and opioid pain medications within 3 months after hospital discharge and had worse McGill Sensory Pain Scores. While there has been considerable recent enthusiasm in including percutaneous cryoneurolysis in rib fracture multimodal pain control bundles (15, 17, 23), our findings suggest this enthusiasm may need to be tempered.

Notably, there were no differences between groups with respect to the SF-12, suggesting health-related quality of life was equivalent between groups. By 3 months after injury, both physical and mental health scores were equivalent to the general U.S. population (24). So, in our patient population, patients were generally able to re-achieve a favorable state of health. Yet 1

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month after injury, patients who underwent intervention had worse physical quality and emotional response to pain, suggesting a deleterious intervention effect. By one year, both groups were equivalent and had reobtained a baseline state of health, free of opioids.

It is intriguing that we did not identify a benefit to percutaneous cryoneurolysis in older patients with rib fractures, as compared to patients undergoing elective surgery. Among children undergoing the Nuss procedure for pectus excavatum, intrathoracic cryoneurolysis has largely become standard of care for pain control based on favorable reduction of pain scores and length of stay when compared to historical cohorts (25). Similarly, in a randomized, sham-controlled study of patients undergoing mastectomy, Ilfeld et al demonstrated marked reduction in pain scores and opioid use among patients undergoing cryoneurolysis which persisted out to 1 year (26).

There are several potential explanations for this discrepancy. For patients undergoing elective surgery, pain is primarily due to local tissue damage at the surgical incisions; the nerves themselves should be intact before surgery and cryoneurolysis performed prior to tissue injury can target and block those nerves in a precise fashion (26). However, for patients with rib fractures, pain can arise from mechanical instability of the chest wall, periosteal and pleural tearing, and ongoing micromotion around fractured rib ends, causing somatic and deep mechanical nociceptive pain in addition to neuropathic pain from nerve injury. Additionally, patients with rib fractures frequently experience pain for a period of time ranging from hours to days pre-procedure; the intercostal nerves are already firing at elevated levels prior cryoneurolysis and may already have experienced Wallerian degeneration due to the initial injury. In this context, cryoneurolysis may create a second hit, amplifying axonal loss, demyelination, and neuroma formation in an area already primed for central sensitization (27).

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Injured nerves often upregulate sodium channels and other excitatory receptors in the dorsal route ganglion (28, 29). In this inflammatory *milieu*, intercostal cryoneurolysis could also lead to aberrant reinnervation or ectopic neuronal discharges, potentially leading to worse neuropathic pain (28, 29).

There are also several potential explanations why no benefit was observed. Cryoneurolysis probes were placed into intercostal spaces, but it is possible probes were not immediately adjacent to the nerves. It is also possible a longer freeze cycle with higher gas flow could improve pain control (30, 31). Second, patients with more severe chest wall injury were excluded as these patients are often treated operatively at our institution. Third, it is possible our time to intervention was too long. Among patients with operatively treated rib fractures, improved outcomes are seen with shorter time to pain control (32). Finally, it could be that no benefit actually exists with CT-guided cryoneurolysis as a pain control modality.

There are several limitations to this study. First, this study was performed in a single ACS Level 1 trauma center; our findings may not be extrapolatable to other contexts. Second, patients, the treatment team, and intervention team were not blinded which introduces bias. Additionally, given this was an intervention-based study placebo effect likely exists which could artificially increase any perceived benefit of cryoneurolysis (33). Third, our study only included adults ≥65 years, which limits interpretation of CT-guided cryoneurolysis technology in a younger population. Fourth, given patients required prone positioning and sedation, it biases our study population to a healthier population. Fifth, we relied on CT-guidance for application of the percutaneous cryoneurolysis, it is possible that other visualization guidance methods, such as ultrasound, may be associated with different results. Sixth, while most patients received cryoneurolysis at 5-6 levels, a few received only 3-4, which may dilute the efficacy. Seventh,

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physical exam not performed at follow-up, so we were unable to assess dermatomal distributions of pain. Eighth, highest pain with deep inspiration at discharge may not capture peak or clinically relevant pain trajectories, and can be inherently subjective (34). Ninth, in order to ensure the clinical trial was pragmatic and to decrease trial burden on institutional resources, we relied on standardized nursing assessment of pain with at least 500cc on incentive spirometry, aligned with our institutional process. However, this could introduce bias if more intense physical activity, elicited by PT or OT, was present for a given patient in a given arm prior to pain score acquisition. Finally, while SF-12 and McGill pain scores were included, these are generic tools and not chest-wall specific. A rib fracture-specific PRO might have been more sensitive in capturing chest wall-specific morbidity.

Conclusion:

CT-guided percutaneous cryoneurolysis did not decrease the pain associated with rib fractures and may not be of benefit when incorporated into a multimodal pain control bundle for older adults. Percutaneous cryoneurolysis may be associated with worse physical quality and emotional response to pain resulting in an increased use of pain medications within the first month after hospital discharge.

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Author Contribution: JDF – study design, literature search, data interpretation, writing, critical revision; AA- data interpretation, critical revision, trial management. ME-data analysis, data

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interpretation, critical revision, RF-data collection, SGB-data collection, JK data analysis, data interpretation, SCL: data collection, AM data collection, JC – study design, KB – study design, SW data interpretation, critical revision, AK-interventional procedure, data interpretation, critical revision, AV-interventional procedure, data interpretation, critical revision, AP-interventional procedure, data interpretation, critical revision, NK- interventional procedure, study design, data interpretation, critical revision.

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Figures:

Figure 1 – CONSORT Flow diagram for patient inclusion into study. ITT, intention to treat; LTFU, long-term follow-up; PI, principal investigator; SOC, standard of care.

Figure 2 - Pain from pre-procedure to 12-month follow-up. Mean pain score with error bars representing 95% CI for (**A**) Pain score during hospitalization, (**B**) McGill Present Pain Intensity (PPI) Score, (**C**) McGill Sensory Pain Score, and (**D**) McGill Affective Pain Score.

Figure 3 - Opioid use from pre-procedure to 12-month follow-up. Proportion of patients on any opioid. Error bars indicate 95% CI. MME, milligram morphine equivalents.

Précis:

CT-guided percutaneous cryoneurolysis did not decrease pain associated with rib fracture in older adults when added to a multimodal pain control bundle and may be associated with worse physical quality and emotional response to pain within 1 month after hospital discharge.

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Table 1 – Demographic, Injury, Chest-Specific Injury, Comorbidity, Pre-Enrollment Pain Therapy and Pre-Enrollment Pain Measures by Group

Characteristic	**Intervention (n = 56)**	**Standard of care (n = 54)**
Demographic		
Age, y, median (IQR)	77 (70-84)	81 (73-87)
Sex, f, n (%)	28 (50)	31 (57)
COPD or asthma, n (%)	15 (27)	4 (7)
BMI >30 kg/m^2, n (%)	3 (5)	6 (11)
Current smoker, n (%)	2 (4)	0 (0)
Diabetes, n (%)	12 (21)	10 (19)
Prior rib fracture, n (%)	10 (18)	7 (13)
Injury		
Mechanism of injury, n (%)		
Fall	39 (70)	43 (80)
Motor vehicle	7 (13)	7 (13)
Motorcycle	2 (4)	2 (4)
Bicycle	6 (11)	2 (4)
Pedestrian vs auto	1 (2)	0 (0)
Other	1 (2)	0 (0)
ISS, median (IQR)	10 (9-14)	10 (9-14)
Admission GCS, median (IQR)	15 (15-15)	15 (15-15)
Intracranial hemorrhage present, n (%)	2 (4)	2 (4)
Facial fracture present, n (%)	2 (4)	2 (4)
Spine fracture present, n (%)	3 (5)	9 (17)
Pelvic fracture present, n (%)	2 (4)	5 (9)
Long bone fracture present, n (%)	2 (4)	6 (11)
Solid organ injury present, n (%)	0 (0)	3 (6)
More than one additional injury present, n (%)	0 (0)	7 (13)
Chest wall-specific injury		
Total no. of rib fractures, median (IQR)	5 (3-6)	5 (3-6)
Laterality of rib fracture, n (%)		
Left only	28 (50)	25 (46)
Right only	21 (38)	23 (43)
Bilateral	7 (13)	6 (11)
Position of fracture, n (%)*		
Anterior	28 (50)	18 (33)
Lateral	28 (50)	25 (46)
Posterior	13 (23)	21 (39)

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Severity of rib fracture, n (%)*		
Undisplaced	29 (52)	24 (44)
Offset	2 (4)	0 (0)
Displaced	32 (57)	39 (72)
Radiographic flail segment present, n (%)	2 (4)	1 (2)
Pneumothorax present, n (%)	12 (21)	13 (24)
Hemothorax present, n (%)	17 (30)	17 (31)
Clavicle fracture present, n (%)	5 (9)	10 (19)
Scapular fracture present, n (%)	4 (7)	0 (0)
Pre-enrollment variables		
Daily opioid equivalents in first 24 h of admission, median (IQR)	25 (15-41)	25 (8-47)
Continuous intercostal nerve block, n (%)	1 (2)	2 (4)
Lidocaine GTT, n (%)	4 (7)	6 (11)
Lidocaine patch, n (%)	47 (84)	51 (94)
Gabapentin, n (%)	31 (55)	41 (76)
NSAID, n (%)	8 (14)	6 (11)
Acetaminophen, n (%)	49 (88)	50 (93)
Highest pain score in first 24 h, median (IQR)	8 (7-10)	8 (7-10)
Cough, n (%)		
None	18 (36)	16 (34)
Weak	12 (24)	16 (34)
Moderate	15 (30)	13 (28)
Strong	5 (10)	2 (4)
Missing	6	7
Incentive spirometry, mL, median (IQR)	1000 (500-1500)	1000 (500-1500)

Intention-to-treat data presented.

*Based on patient having at least 1 fracture with this location or morphology.

GCS, Glasgow Coma Scale; GTT, continuous infusion; IQR, interquartile range; ISS, Injury Severity Score.

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Table 2 – Hospitalization and Discharge Destination by Group

Characteristic	**Intervention (n=56)**	**Standard of care (n=54)**	**p Value**
Hospitalization variable			
Hospital LOS, d, median (IQR)	4 (3,7)	6 (3,7)	0.77
Admission to ICU, n (%)	6 (11)	6 (11)	0.95
Discharge variable			
Gabapentin prescribed, n (%)	40 (71)	42 (78)	0.51
Lidocaine patch prescribed, n (%)	44 (79)	48 (89)	0.20
NSAID prescribed, n (%)	11 (20)	3 (6)	0.03
Acetaminophen prescribed, n (%)	52 (93)	51 (94)	>0.99
Highest pain score in 24 h before discharge, mean (SD)	5.6 (2.4)	5.9 (2.4)	0.49
Oral opioid usage on day of discharge > 0 MME, n (%)	46 (82)	46 (85)	0.80
Oral MME on day of discharge, median (IQR)	24 (6-45)	15 (4-34)	0.29

Intention-to-treat data presented.

IQR, interquartile range; LOS, length of stay; MME, milligram morphine equivalents; NSAID, nonsteroidal anti-inflammatory.

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Table 3 – Pain Medication Use and Patient-Reported Outcomes at 1-, 3- and 12-Month Follow-Up

Characteristic	Intervention (n=56)	SOC (n=54)	p Value
1-mo follow-up, n	39	38	
Gabapentin still used, n (%)*	25 (45)	6 (12)	<0.001
Lidocaine patch still used, n (%)*	24 (44)	12 (24)	0.04
NSAIDs used, n (%)*	13 (24)	12 (24)	0.97
Acetaminophen still used, n (%)*	37 (67)	34 (68)	0.94
McGill Pain Score: PPI, mean (SD)†	2.7 (2.4)	1.7 (1.8)	0.04
McGill Pain Score: Sensory, mean (SD)†	4.6 (4.9)	1.5 (2.0)	<0.001
McGill Pain Score: Affective, median (IQR)†	1 (0-2)	0 (0-0)	0.01
SF-12 PCS, median (IQR)‡	32.9 (25.2-41.5)	29.9 (24.2-43.7)	0.96
SF-12 MCS, median (IQR)‡	55.1 (49.7-58.1)	55.7 (46.1-60.1)	0.72
Using oral opioids, n (%)§	18 (32)	6 (12)	0.01
Daily total MME, median (IQR)§	0 (0-5)	0 (0-0)	0.01
3-mo follow-up	35	37	
Gabapentin used, n (%)*	12 (23)	6 (13)	0.20
Lidocaine patch used, n (%)*	14 (26)	3 (6)	0.008
NSAIDs used, n (%)*	12 (23)	5 (11)	0.11
Acetaminophen used, n (%)*	21 (40)	18 (38)	0.89
McGill Pain Score: PPI, mean (SD)†	1.4 (1.8)	0.9 (1.3)	0.15
McGill Pain Score: Sensory, mean (SD)†	1.0 (1.4)	0.6 (1.7)	0.32
McGill Pain Score: Affective, median (IQR)†	0 (0-0)	0 (0-0)	0.70
SF-12 PCS, median (IQR)‡	51.4 (43.0-53.6)	48.0 (29.9-55.5)	0.36
SF-12 MCS, median (IQR)‡	58.7 (53.1-60.8)	57.8 (49.4-60.3)	0.38
Using oral opioids, n (%)§	4 (8)	1 (2)	0.37

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Daily total MME, median (IQR)[§]	0 (0-0)	0 (0-0)	0.20
12-mo follow-up, n	30	32	
Gabapentin used, n (%)*	9 (18)	5 (11)	0.37
Lidocaine patch used, n (%)*	5 (10)	2 (4)	0.44
NSAIDs used, n (%)*	8 (16)	8 (18)	0.78
Acetaminophen used, n (%)*	13 (25)	16 (36)	0.28
McGill Pain Score: PPI, mean (SD)[†]	0.6 (1.1)	0.6 (1.3)	0.87
McGill Pain Score: Sensory, mean (SD)[†]	0.4 (0.7)	0.3 (0.7)	0.68
McGill Pain Score: Affective, median (IQR)[†]	0 (0-0)	0 (0-0)	>0.99
SF-12 PCS, median (IQR)[‡]	52.8 (47.9-55.5)	51.5 (44.8-55.5)	0.33
SF-12 MCS, median (IQR)[‡]	57.8 (53.2-60.7)	59.7 (55.9-61.6)	0.07

Intention to treat data presented.

[*]Data for non-opioid pain control were taken from charts even if patients did not follow up. Sample sizes after excluding missing data are as follows: 1-month follow-up: n = 55 for intervention, n = 50 for SOC; (n = 49 for lidocaine); 3-month follow-up: n = 53 for intervention, n = 47 for SoC; 12-month follow-up: n = 51, n = 45 for SoC.

[†]Due to missing data, not all completed follow-ups have McGill scores. Actual sample sizes are as follows: 1-month follow-up: n = 39 for intervention, n = 37 (PPI), n = 36 (sensory), n = 34 (affective) for SOC; 3-month follow-up: n = 33 (PPI and affective), n = 34 (sensory) for intervention, n = 35 (PPI and sensory), n= 33 (affective) for SOC; 12-month follow-up: n = 30 for intervention, n = 31 for SOC.

[‡]Due to missing data, not all completed follow-ups have SF-12 scores. Actual sample sizes are as follows: 1-month follow-up: n = 39 for intervention, n = 36 for SOC; 3-month follow-up: n = 35 for intervention, n = 34 for SOC; 12-month follow-up: n = 29 for intervention, n = 31 for SOC.

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[§]Data for MME were taken from charts even if patients did not follow up. Sample sizes after excluding missing data are as follows: 1-month follow-up: n = 56 for intervention, n = 51 for SOC; 3-month follow-up: n = 53 for intervention, n = 48 for SoC.

IQR, interquartile range; MCS, mental component score; MME, milligram morphine equivalents; N/A, not applicable; NSAID, nonsteroidal anti-inflammatory; PCS, physical component score; PPI, present pain intensity; SOC, standard of care.

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Figure 1

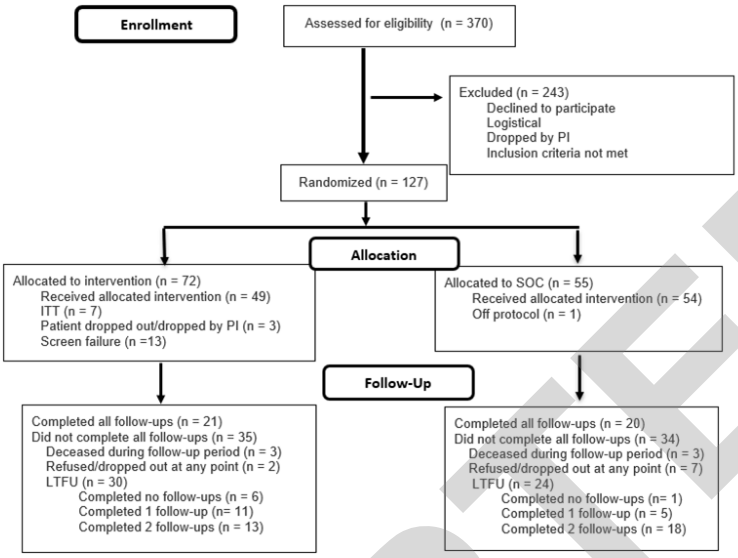

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Figure2

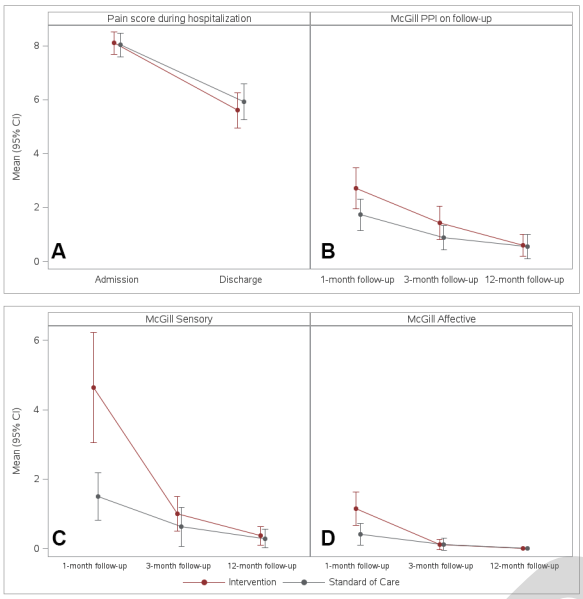

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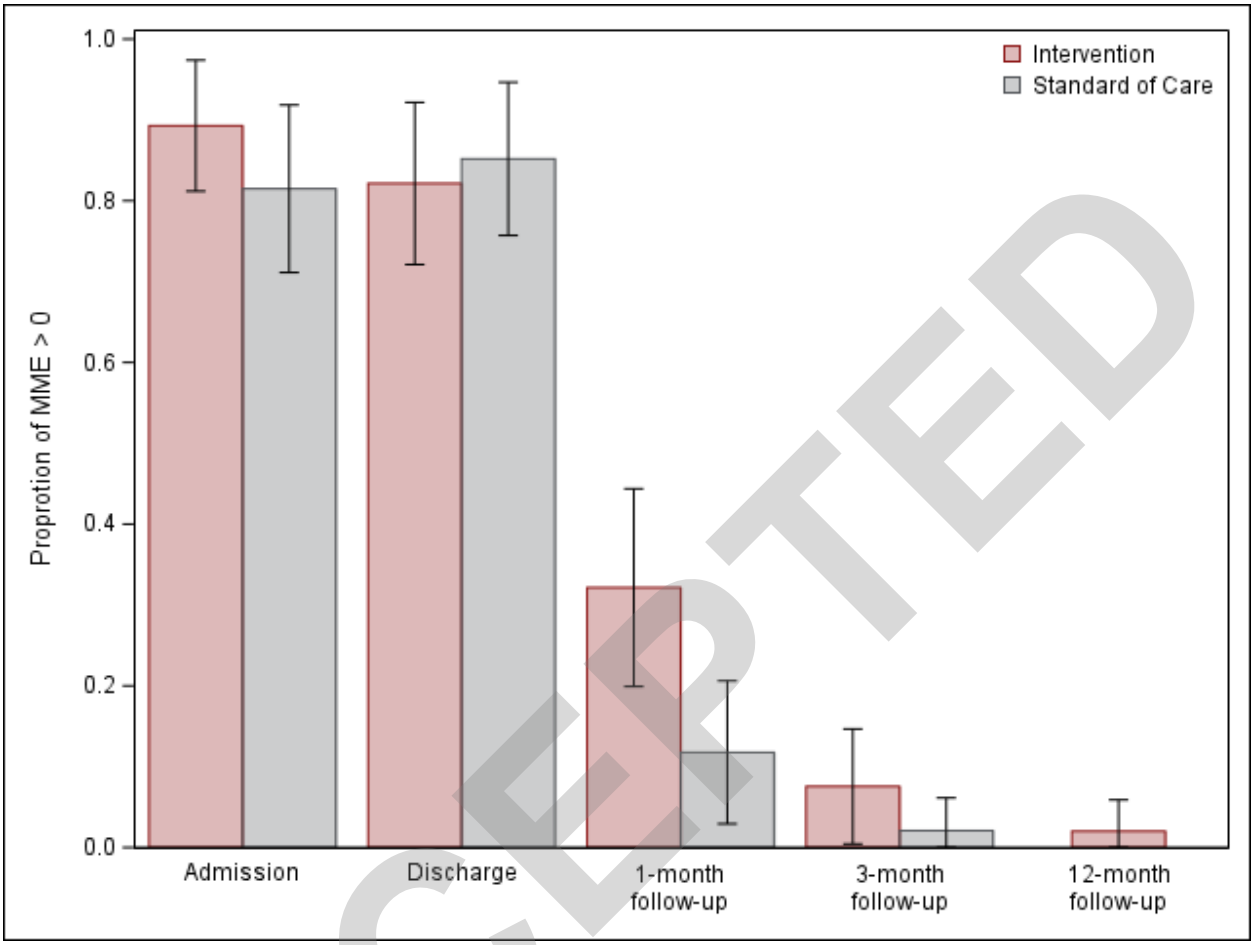
Figure 3



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